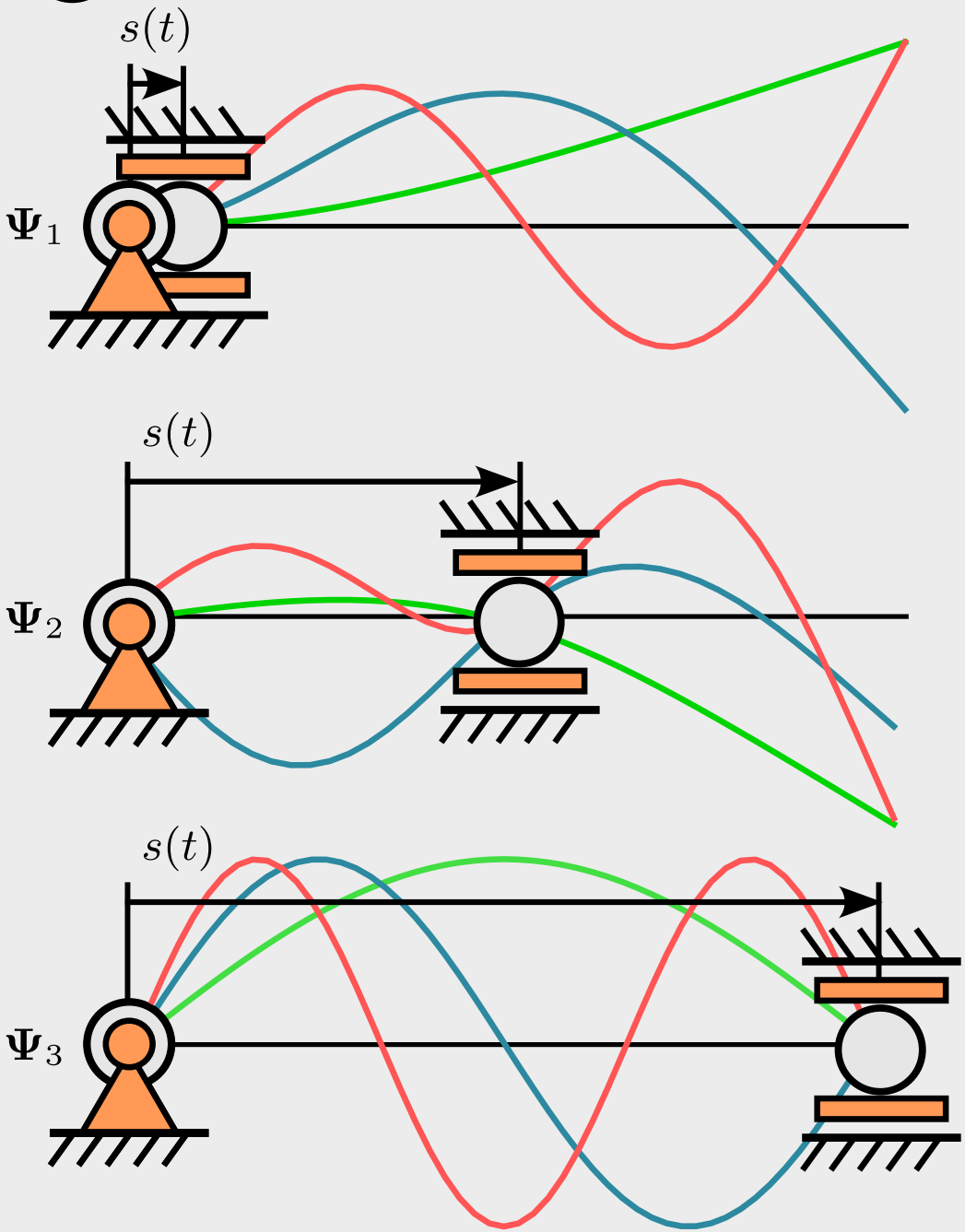
1

collect snapshots, e.g.:

$n_{\text{snap.}} = 3$ and $r_{\text{snap.}} = 3$



2

assemble *pool*

$$\Psi_{\text{all}} = [\Psi_1, \Psi_2, \Psi_3]$$

$$\Psi_{\text{all}} \in \mathbb{R}^{n_{\text{DOF}} \times (n_{\text{snap.}} r_{\text{snap.}})}$$

3

perform SVD

$$U \Sigma V^* = \text{SVD}\left(\Psi_{\text{all}}\right)$$

4

select global modes

$$\Psi_{\text{glob.}} = U_{:,1:r_{\text{glob.}}}$$

$$\Psi_{\text{glob.}} \in \mathbb{R}^{n_{\text{DOF}} \times r_{\text{glob.}}}$$

5

reduce system (const. matrices)

$$M_{\text{red.}} = \Psi_{\text{glob.}}^{\text{T}} M \Psi_{\text{glob.}}$$

$$K_{\text{red.}} = \Psi_{\text{glob.}}^{\text{T}} K \Psi_{\text{glob.}}$$

6

set up Lagrangians Type 1

$$f_{\text{red.}}(t) = \Psi_{\text{glob.}}^{\text{T}} f(t)$$

$$C_{q,\text{red}}\left(s(t)\right) = C_q\left(s(t)\right) \Psi_{\text{glob}}$$

$$g\left(s(t), q(t)\right) = 0$$

$$M_{\text{red.}} \ddot{p}(t) + K_{\text{red.}} p(t) + C_{\text{q,red.}}^{\text{T}}\left(s(t)\right) \lambda(t) = f_{\text{red.}}(t)$$